2026 ASME STUDENT MECHANISM & ROBOT DESIGN COMPETITION

A Cable-Driven Multi-modal Rolling Robot Based on Twin Waterbomb 3-RSR Origami Parallel Mechanism

TENDON PROGRAM → COMPLIANT BODY DEFORMATION → ELASTIC ENERGY / TENDON FORCE → CONTACT-CONSTRAINT SELECTION → GAIT

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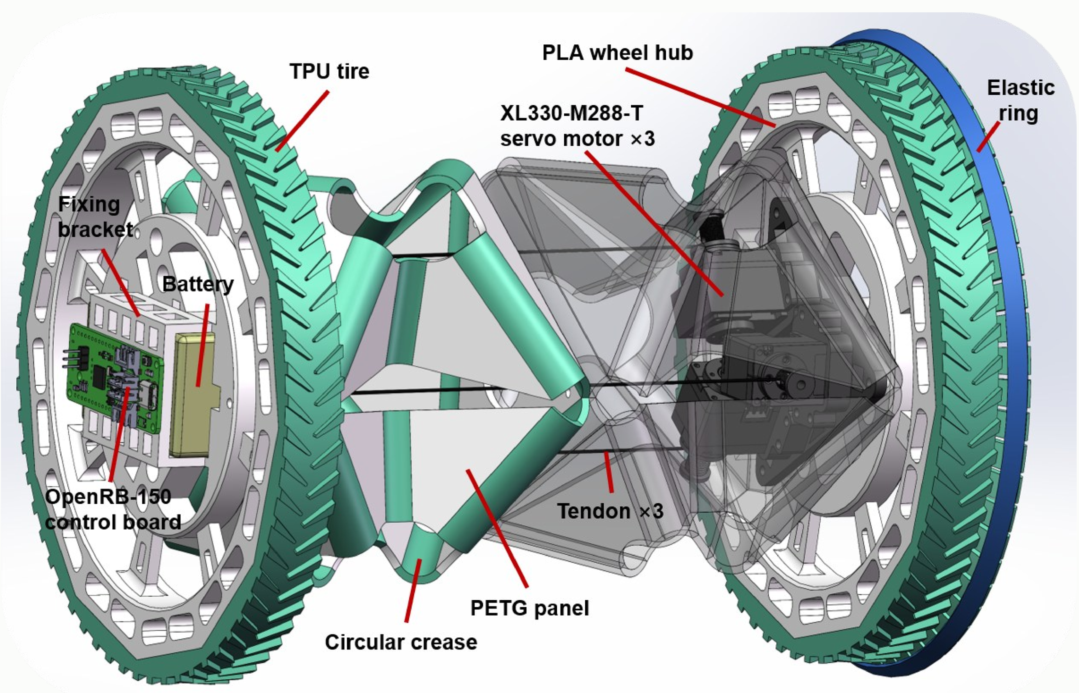
DESIGN CONTRIBUTION

1. Multimaterial FDM enables assembly-free fabrication;
2. A compliant Waterbomb origami body serves as the rolling robot torso;
3. A contact-constraint selection framework enables locomotion-mode switching without hardware changes.

1 STRUCTURE, KINEMATICS & PROTOTYPE

Configuration : a compliant Waterbomb origami body serves as the deformable torso of the rolling robot.

1 · COMPLIANT BODY AS ROBOT TORSO



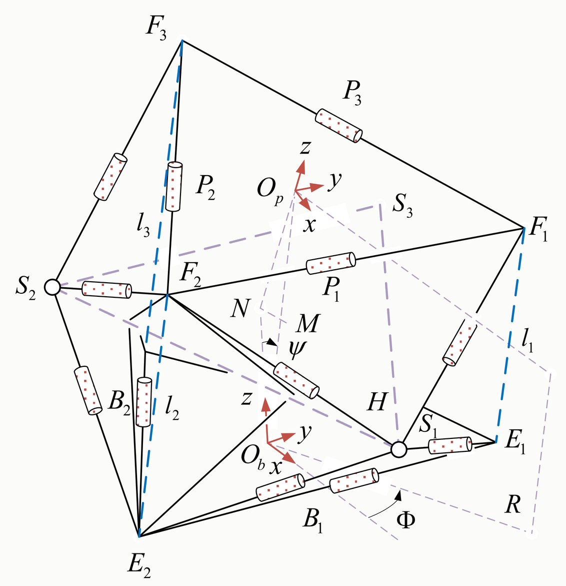
DESIGN RESPONSE

- compliant Waterbomb body forms the robot torso
- twin 3-RSR modules share three tendons
- tendon input programs the 1T2R body pose
- rubber elastic rings and mass bias shape contact

RESULT

One body produces roll, turn and crawl through phase, amplitude and DC offset programming.

ANALYTICAL TENDON-TO-POSE MODEL



The Waterbomb 3-RSR kinematics maps the three tendon lengths to axial shortening and differential-length components governing tilt and precession.

ANALYTICAL TENDON-TO-POSE MAP

$$q = [r_0, \psi, \Phi]^T, \quad DOF = 3\ (1T2R)$$

$$\ell_i = r_0 - 4r sin(\psi) cos(\varphi_i + \frac{\pi}{3} - \Phi)$$

$$s_i = cn_i, \quad \ell_i^{tot} = 2L_0 - s_i$$

- the spool map converts motor turns to tendon take-up;
- three equally spaced R-S-R branches map limb lengths to axial translation, nutation and azimuth;
- each tendon passes through both modules, coupling their 1T2R response.

COMMON / DIFFERENTIAL CONTROL COORDINATES

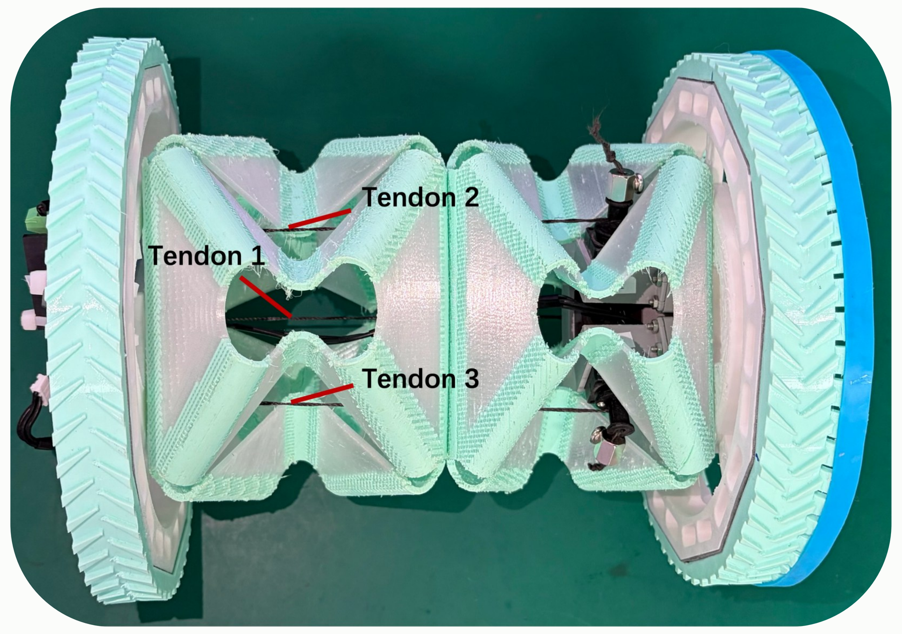
$$\bar{s} = \frac{s_1 + s_2 + s_3}{3}, \quad \tilde{s}_i = s_i - \bar{s}, \quad \sum_{i=1}^{3} \tilde{s}_i = 0$$

$$r_0 \approx L_0 - \frac{\bar{s}}{2}, \quad \ell_i - r_0 \approx -\frac{\tilde{s}_i}{2}$$

common mode → axial shortening | zero-sum component → tilt / precession

With two identical modules on each through-tendon, the measured take-up is shared equally: $\bar{s}/2$ sets each module's axial shortening, while $\tilde{s}_i/2$ sets its differential tilt and precession input.

ONE FIXED-HARDWARE PROTOTYPE REALIZES THREE GAITS



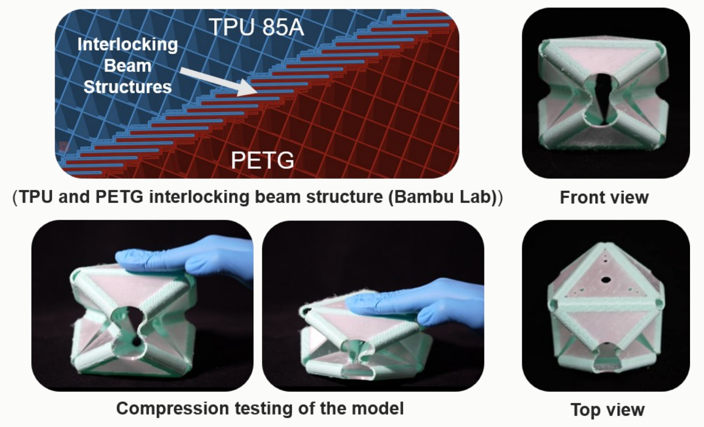
The prototype realizes the shared-tendon structure; its geometry and mass bias inform the contact analysis for M1, M2 and M3.

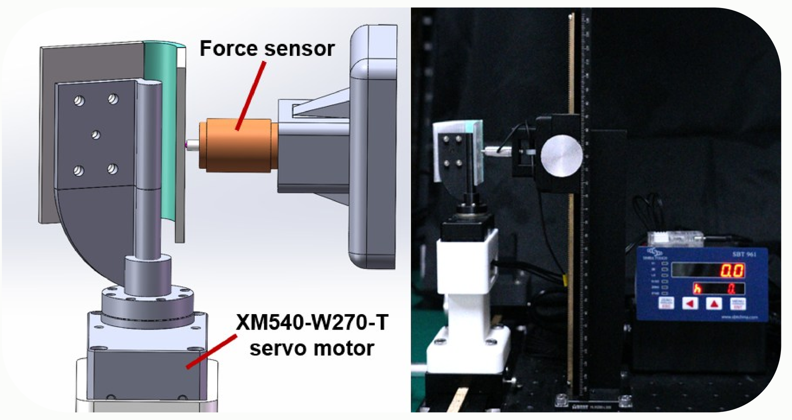
879 g	230 mm	+15.03 g	3
TOTAL MASS	WHEEL SPACING	MOTOR-SIDE BIAS	SHARED TENDONS

2 FABRICATION, EXPERIMENTAL & VALIDATION

Manufacturing : multimaterial FDM joins rigid PETG facets and compliant TPU creases into an assembly-free body; calibration and compression validate its mechanics.

BUILD → CALIBRATE → MODEL → MEASURE → COMPARE





2 · ASSEMBLY-FREE FABRICATION

Transparent PETG facets and TPU 85A facets are joined by tooth-shaped mechanical interlocks in a 0.20 mm FDM process. The flexible TPU connections form the creases characterized by torsion testing.

CALIBRATED CREASE-TO-FORCE MODEL

A 0 to 40° sweep with 200 Hz force sensing identifies rotational stiffness over the fitted 5 to 35° interval. The measured stiffness parameterizes the energy model.

THEORY CHAIN: CREASE ENERGY → TENDON FORCE

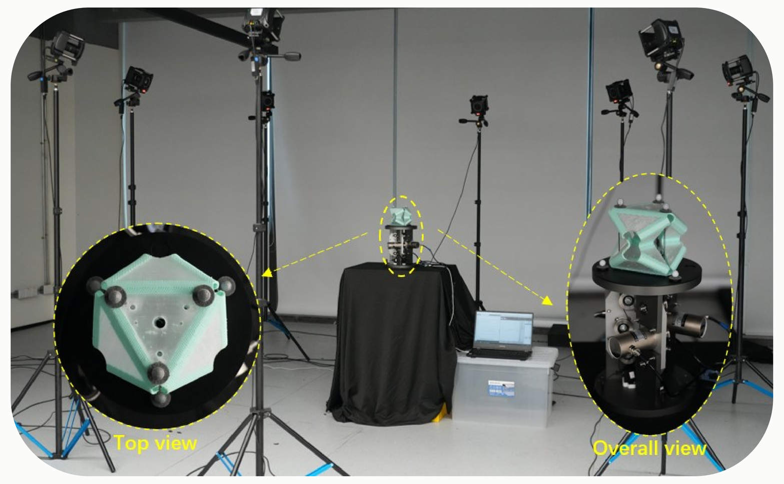
$$K_{\theta,j} = k_{\ell,j} \cdot \ell_{c,j}, \quad K_{\theta,beam} = \frac{EI}{L}, \quad I = \frac{bh^3}{12}$$

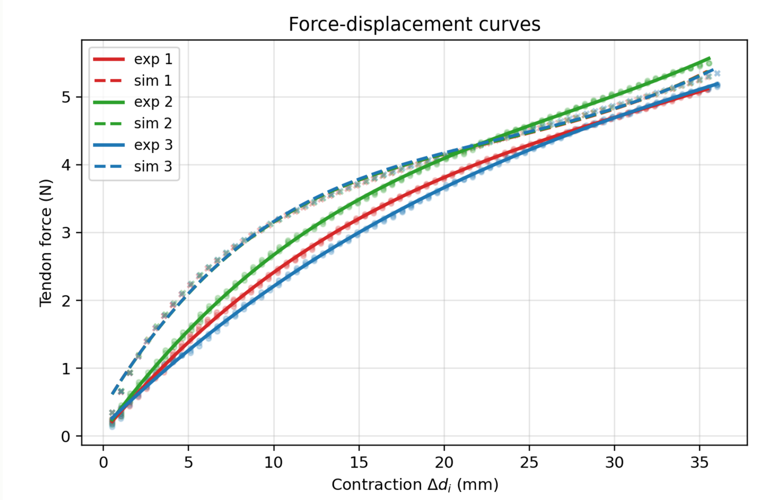
$$U(\ell) = \frac{1}{2}\sum_{j=1}^{24} k_{\ell,j}\ell_{c,j}[\alpha_j(\ell) - \alpha_{j0}]^2$$

$$T_i = -\frac{\partial U}{\partial \ell_i}$$

$K_{\theta,exp}$ = 0.0410 N·m·rad⁻¹ | R^2 = 0.9941 | 91.1% of the beam estimate

The identified crease stiffness completes the quasi-static energy-to-force model.
Synchronous axial compression provides the simplest direct validation of the predicted tendon-force response.





MEASURED TENDON LENGTH + FORCE

The NOKOV motion-capture system reconstructs tendon-node trajectories at 100 Hz, while three force channels sample at 5 Hz. Marker-derived tendon length reduces model-input uncertainty from guide-hole routing and servo position-loop error.

PREDICTED FORCE VS. MEASURED FORCE

The synchronized test provides measured contraction and tendon force. Evaluating the model at the measured tendon length gives r = 0.986 to 0.995 (mean 0.992); force RMSE T1/T2/T3 = 0.505 / 0.346 / 0.615 N.

The model reproduces the measured force trend across all three tendons, with mean correlation r = 0.992.

MODEL VALIDATION

$$\Delta d_i(t) = d_i(0) - d_i(t), \quad \ell_i(t) = L_0 - \Delta d_i(t)$$

202 aligned samples · stroke 0.51 to 36.03 mm
max sync error 0.922 mm (2.6%) · mean(Tmeas - Tpred) = +0.386 N

$K_\theta \rightarrow U(\ell) \rightarrow T_{pred}(\ell) \leftrightarrow T_{meas}(\ell) \rightarrow$ CONTACT MECHANICS → OBSERVED GAIT

ONE MODEL CHAIN, THREE GAIT MECHANISMS

MEASURED	$K_{\theta,exp}$ · Δd(t) · T_{meas} · three gait image sequences
MODELED	U(ℓ), T_{pred}(ℓ) and contact-mode relations for M1, M2 and M3
GAIT OUTPUT	M1 continuous roll · M2 arc turn · M3 axial crawl

Analytical kinematics, measured crease stiffness and predicted tendon force form one experimentally validated model chain.

THREE INNOVATIONS · ONE INTEGRATED ROBOT SYSTEM

3 CONTACT MECHANICS & MULTIMODAL LOCOMOTION

Mechanism :
The proposed contact-constraint selection framework maps programmed body deformation to M1 rolling, M2 turning and M3 crawling.

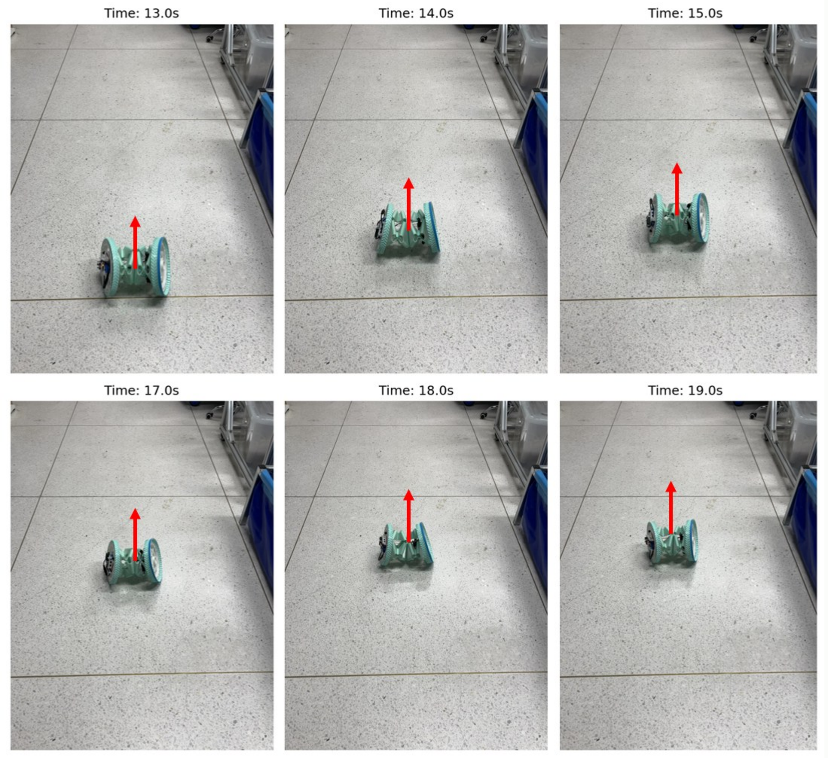
3 · CONTACT-CONSTRAINT SELECTION FRAMEWORK

$$C_i = \mu_i \cdot N_i$$

$$stick: \quad \|f_i^t\| \le C_i, \quad v_i^{rel} = 0$$

$$slip: \quad f_i^t = -C_i \frac{v_i^{rel}}{\|v_i^{rel}\|}$$

M1: paired rolling contact · M2: differential wheel motion · M3: alternating end loading



M1 · CONTINUOUS ROLL

INPUT	equal-amplitude waves with 120° phase offsets
IDEAL	both wheels satisfy no-slip
MODEL	frequency-doubled precession drives rolling

$$\Phi \approx 2\omega$$

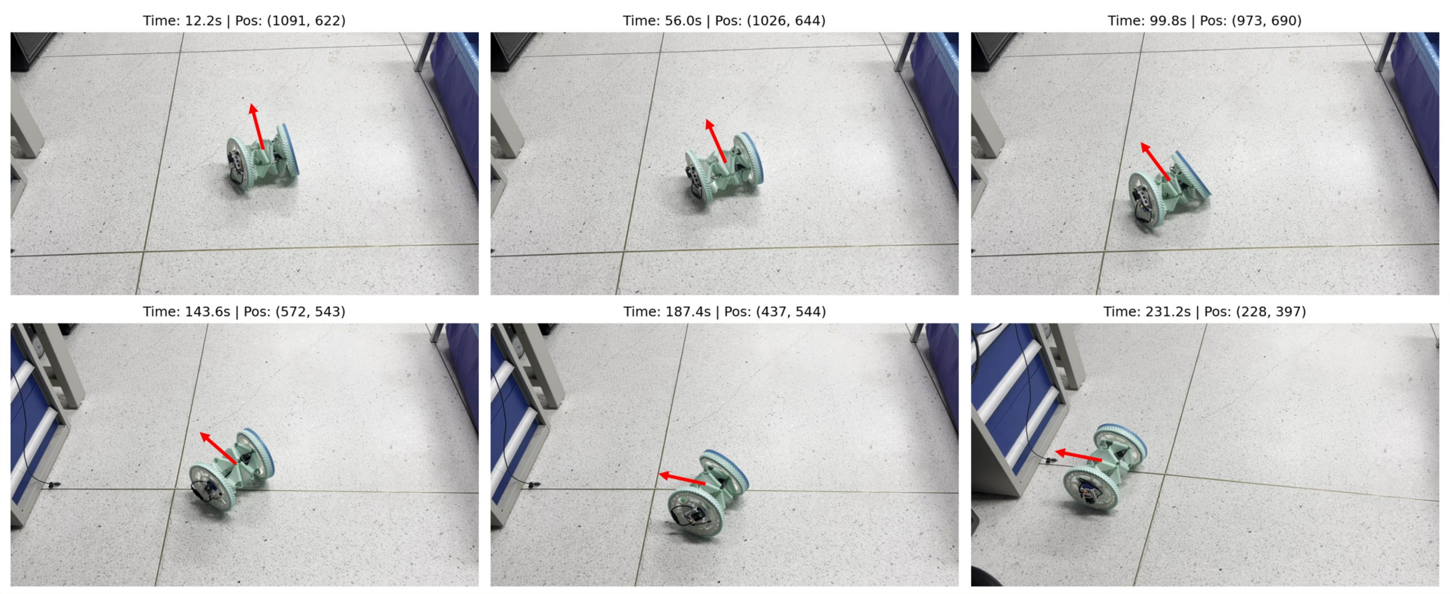
$$\Delta y_M^{pred} \approx 2\pi\langle\rho\rangle \approx 0.594\ m/cycle$$

The 120° phase program produces frequency-doubled precession. With the effective rolling radius, the model predicts 0.594 m per rectified shape cycle. The six-frame sequence shows continuous rolling with lateral drift.

M2 · ARC TURN

$$R_{arc} = \frac{L_w}{2}\left|\frac{v_o + v_i}{v_o - v_i}\right|$$

INPUT · asymmetric amplitude and DC-offset bias. MODEL · v_o and v_i are outer and inner wheel-center speeds in the turning relation. OBSERVED · the image sequence shows an arc turn with yaw ≈ 165° and R ≈ 1 m. Differential wheel speeds set the turning radius.



M3 · AXIAL CRAWL

$$\mu_R^{tilt} \cdot N_R^{(c)} < |F_{ax}^{(c)}| \le \mu_L \cdot N_L^{(c)}$$

$$\mu_L \cdot N_L^{(r)} < |F_{ax}^{(r)}| \le \mu_R^{up} \cdot N_R^{(r)}$$

$$\Delta x_M^{cycle} \le \Delta L_{ax}$$

INPUT · common DC-offset contraction and release. MODEL · the compression and release force windows produce alternating end loading, converting axial deformation into directional motion. OBSERVED · the 0 to 13 s sequence shows net axial crawling.